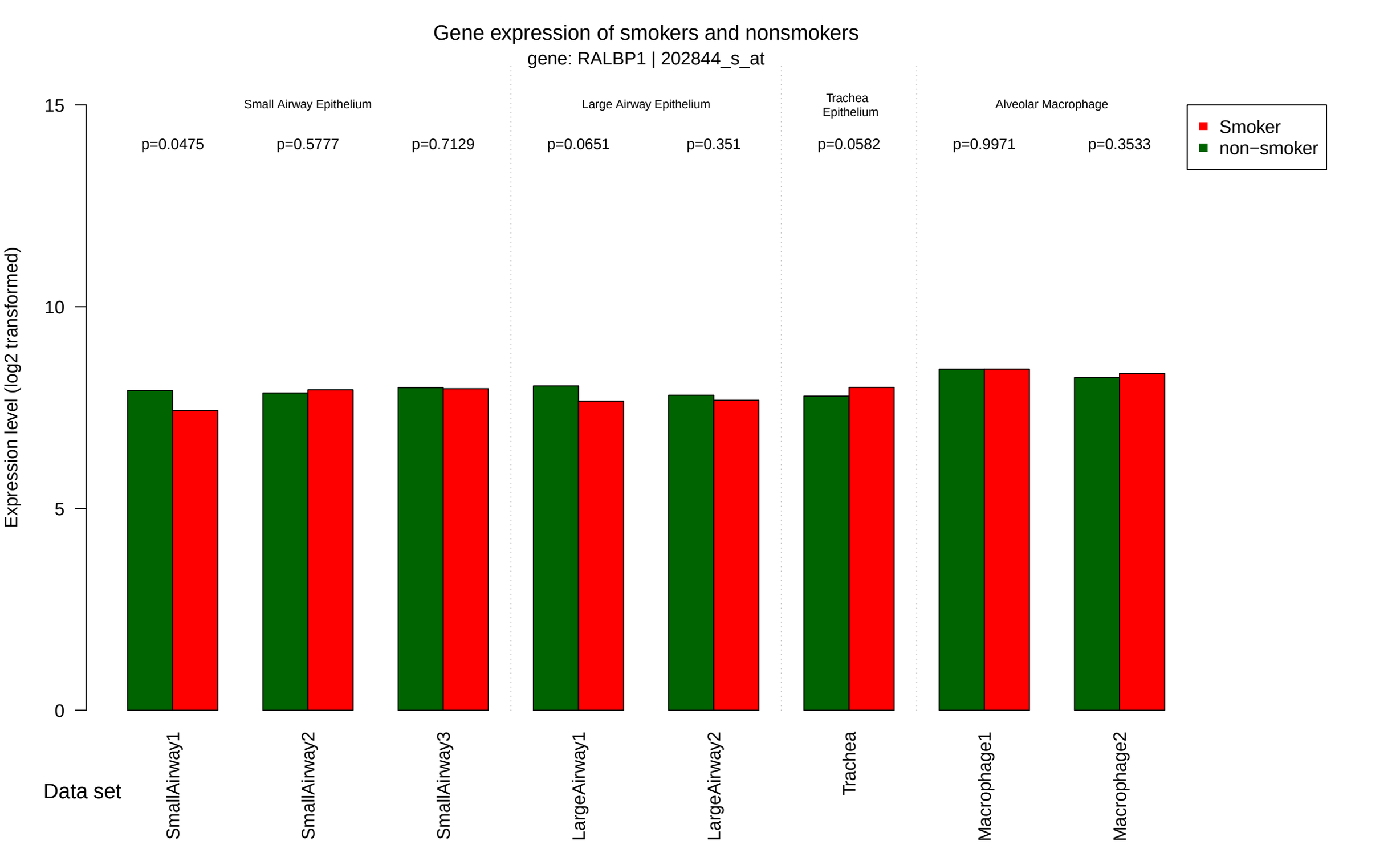
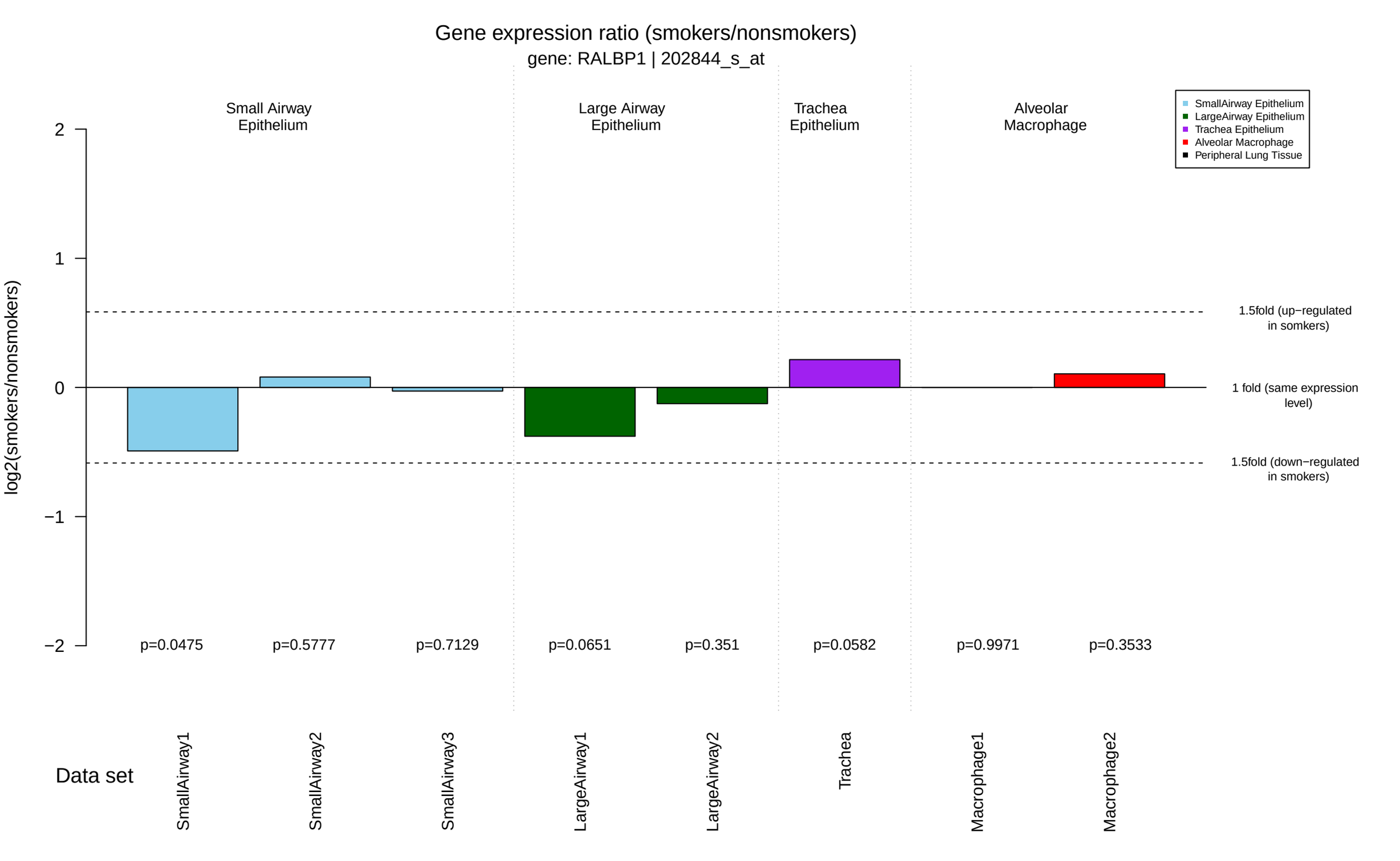
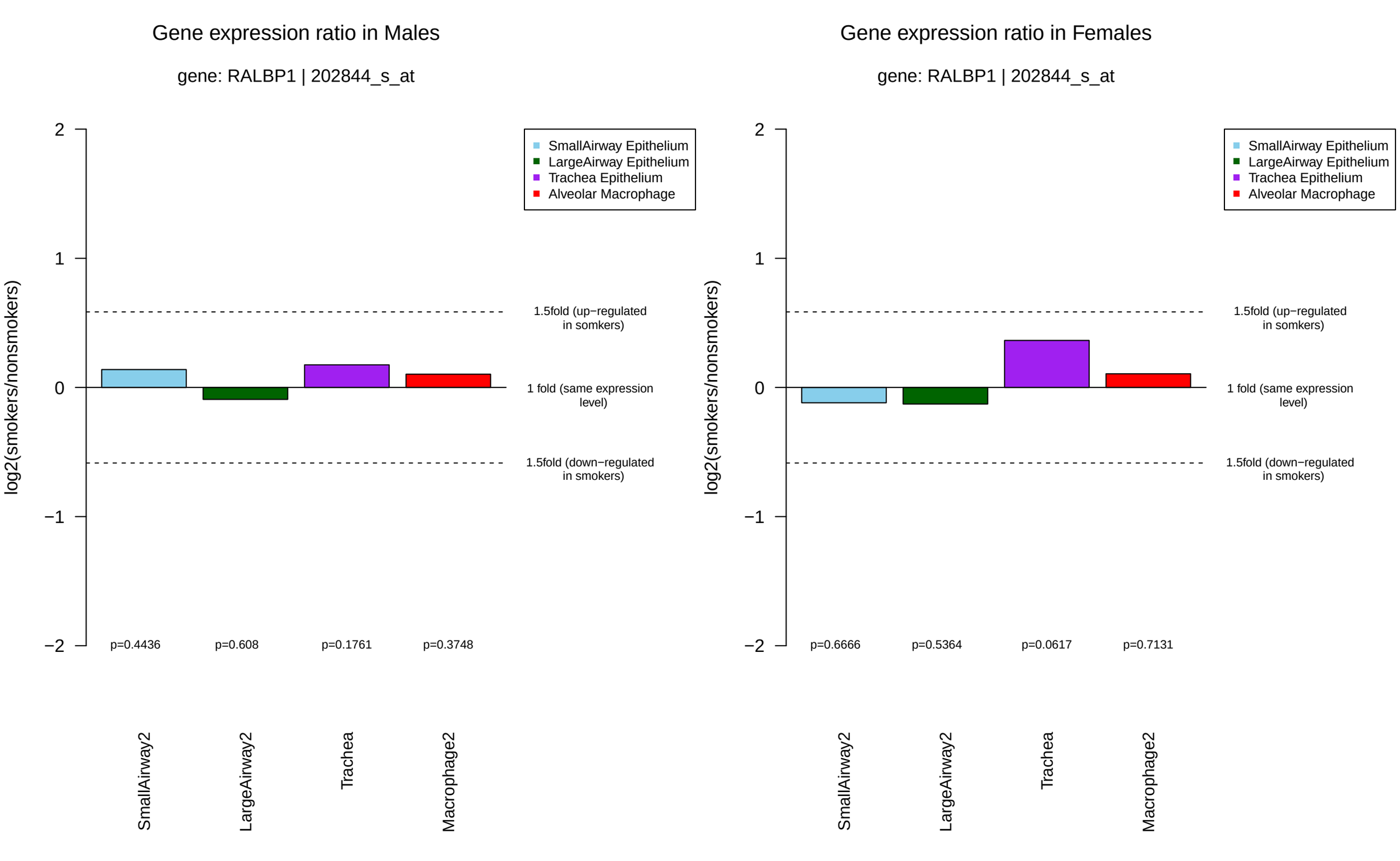
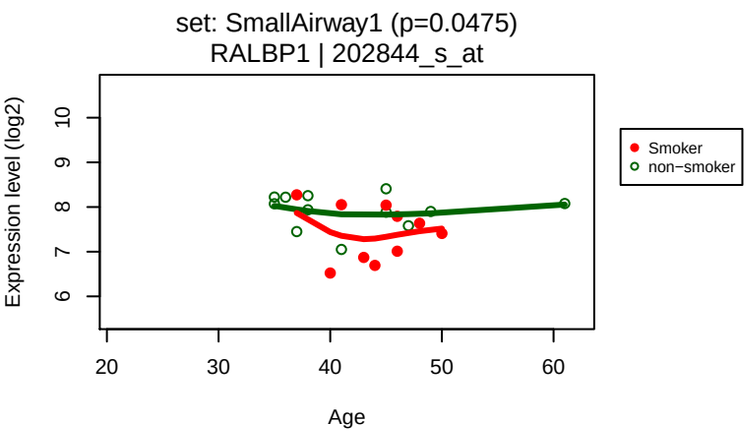
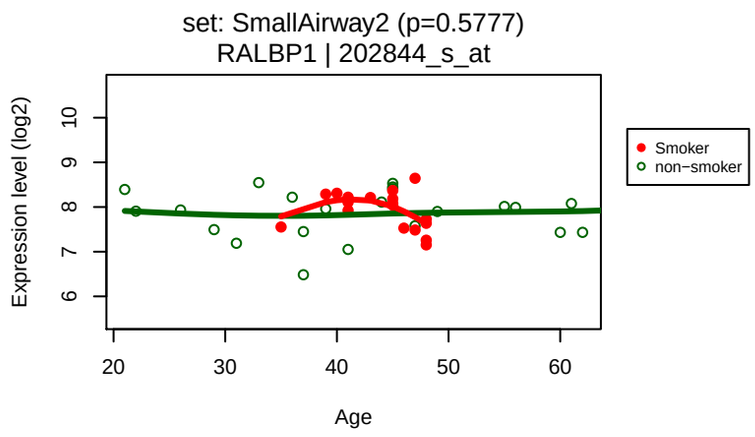
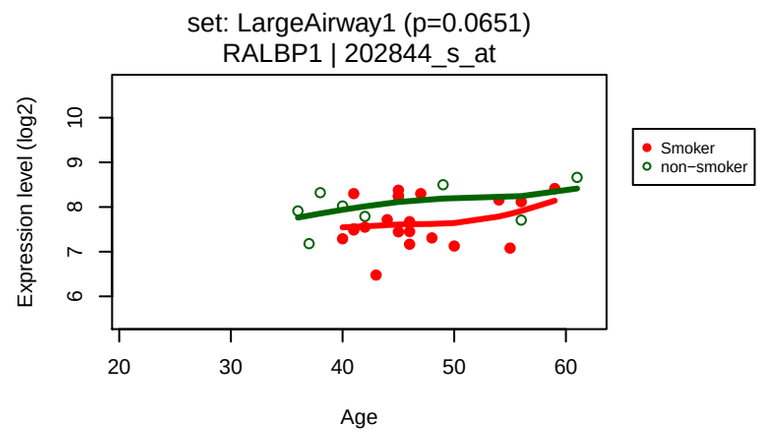

Gene expression of smokers and nonsmokers
gene: RALBP1 | 202844_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0475
p=0.5777
p=0.7129
p=0.0651
p=0.351
p=0.0582
p=0.9971
p=0.3533
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

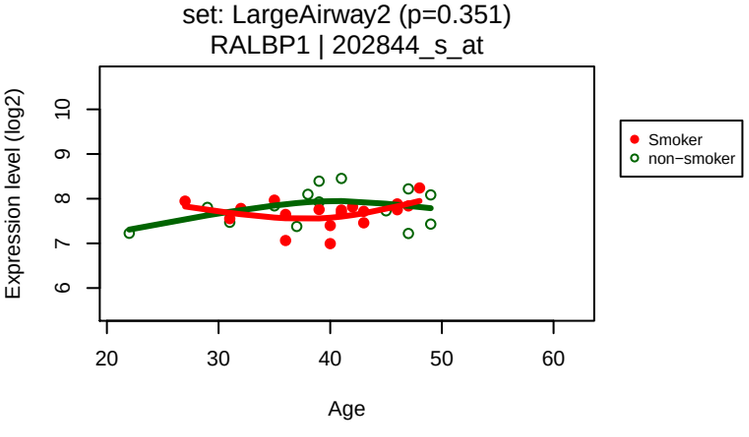

Gene expression ratio (smokers/nonsmokers)
gene: RALBP1 | 202844_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0475
p=0.5777
p=0.7129
p=0.0651
p=0.351
p=0.0582
p=0.9971
p=0.3533
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

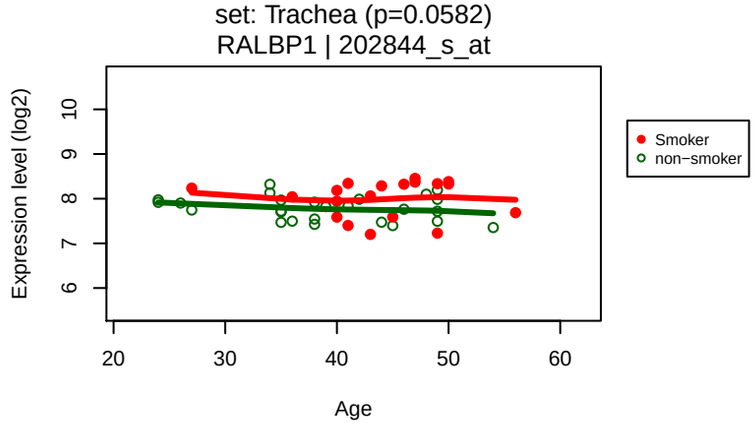

Gene expression ratio in Males
gene: RALBP1 | 202844_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4436
p=0.608
p=0.1761
p=0.3748
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RALBP1 | 202844_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6666
p=0.5364
p=0.0617
p=0.7131
SmallAirway2
LargeAirway2
Trachea
Macrophage2

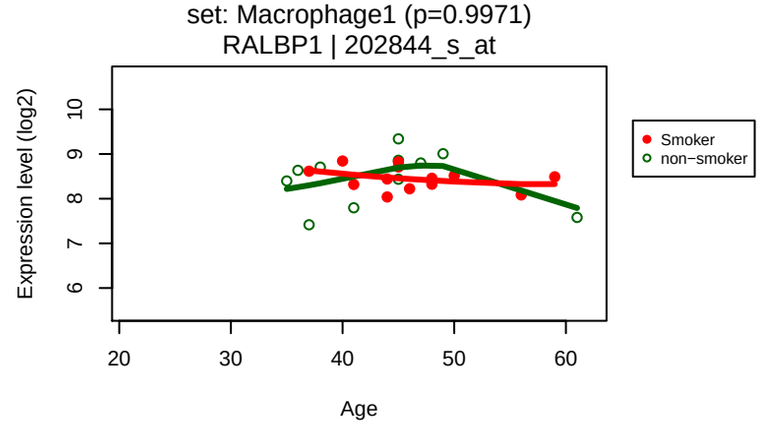

set: SmallAirway1 (p=0.0475)
RALBP1 | 202844_s_at
Expression level (log2)
Age
Smoker
non−smoker

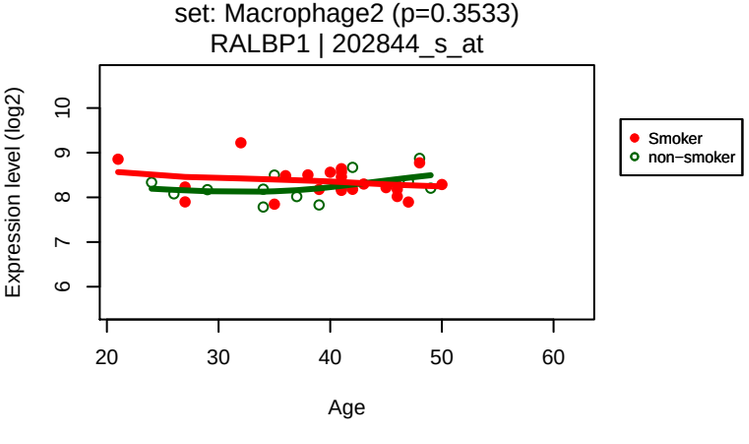

set: SmallAirway2 (p=0.5777)
RALBP1 | 202844_s_at
Expression level (log2)
Age
Smoker
non−smoker

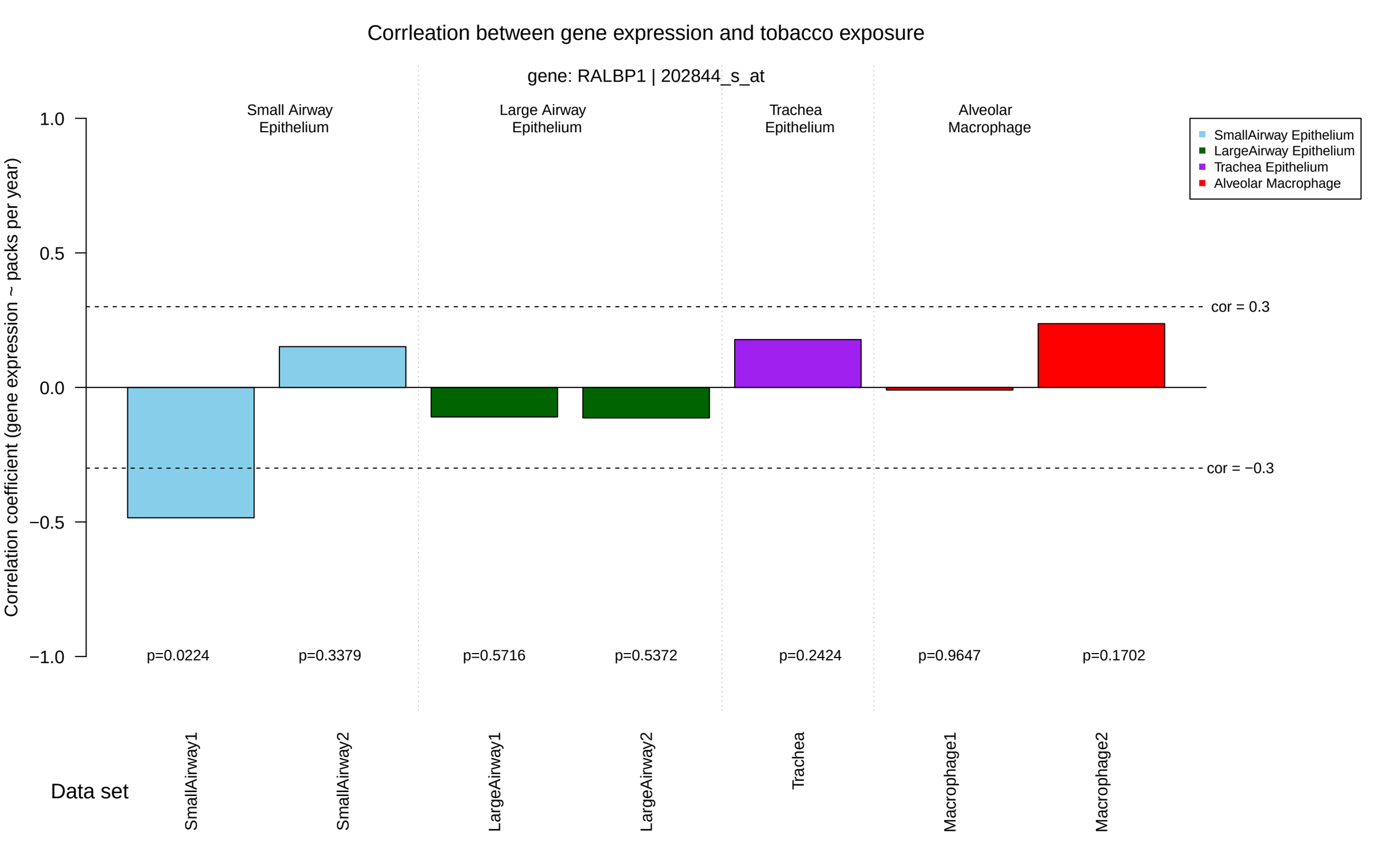

set: LargeAirway1 (p=0.0651)
RALBP1 | 202844_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.351)
RALBP1 | 202844_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0582)
RALBP1 | 202844_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9971)
RALBP1 | 202844_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3533)
RALBP1 | 202844_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RALBP1 | 202844_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0224
p=0.3379
p=0.5716
p=0.5372
p=0.2424
p=0.9647
p=0.1702
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2